

Natural Language Meets Analytics: How IBM Powers Enterprise-Grade Text2SQL

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IBM

Business Insights

Define the data

Projects / Agent Data / Agent Data

Edit enrichment [Enrich all assets](#)

Assets (5) Columns (90) Review key relationships

Find columns (search by column name, asset name, business term, classification, or data class)

Last refreshed: 4/14/25, 1:11 PM

Column	Asset	Display name	Description	Business terms	Data class	Class
agent_count	agent Agent Data... / sample_cat...	agent count	■ RZ The total number of agents associated with...	Actual + 2 more	Boolean	—
agent_email	agent Agent Data... / sample_cat...	agent email	■ RZ The email address of the agent	Target + 2 more	Email Ad...	—
agent_employment_start_date	agent Agent Data... / sample_cat...	agent employment start date	■ RZ The date when the agent began their employment.	Salary + 2 more	Date	View
agent_first_name	agent Agent Data... / sample_cat...	agent first name	■ RZ The first name of the agent	German + 2 more	—	—
agent_highest_education_level	agent Agent Data... / sample_cat...	agent highest education level	■ RZ The highest level of education achieved by th...	Salary + 2 more	Organizat...	—
agent_hourly_pay_rate	agent Agent Data... / sample_cat...	agent hourly pay rate	■ RZ The rate at which the agent is paid per hour.	Salary + 2 more	—	—
agent_id	agent	agent identifier	—	Target	Identifier	1 of 2 pages

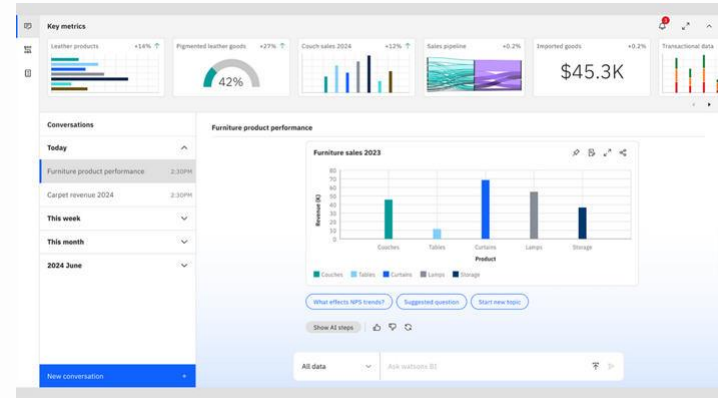
Items per page: 50 1-50 of 90 items

Industry-agnostic
data insights

Realize the full potential
of your data using
Generative AI and
Text2SQL techniques

Discover Insights

- Surface insights in seconds and accelerate decision-making
- Utilize semantic automation and metrics generation to reduce your data prep time
- Get trusted insights based on your business logic and metrics you care about
- Use your existing tools than facing yet another disruption



A Day in the Life



Satish B.
Data Steward

Main Responsibilities:

- ❑ *Governs and ensures data quality, security, and compliance.*
- ❑ *Manages access, curates, and publishes data products.*
- ❑ *Supports business users by handling queries and requests.*

• Before Text2SQL

- Inbox overwhelmed with one-off requests for data pulls and query support.
- Buried in repetitive SQL writing, ticket management, and back-and-forth clarifications.
- Backlog continues to pile up, leaving little capacity for governance or strategic work.

• After Text2SQL

- Reviews usage logs and monitoring automated query activity - no flood of requests
- Better visibility into data needs of the enterprise
- Invests time in governance, data quality, and enhancing data products.



Ganesh R.
LOB User

Main Responsibilities:

- ❑ *Asks business questions and generate insights from data.*
- ❑ *Uses insights to inform decisions and drives outcomes.*
- ❑ *Collaborates with stewards to access the right data sources.*

• Before Text2SQL

- A business question arises, but the only option is to email or file a ticket with the data team.
- Work stalls while waiting for a response, often relying on outdated or incomplete reports in the meantime.
- Receives query results, but they raise new questions—triggering yet another request and more delays.

• After Text2SQL

- Quickly types a plain-English query (e.g., “Show churn by region last 6 months”) and gets instant results.
- Refines and explores queries independently—no dependency, no backlog.
- Packages fresh insights into a leadership-ready report the same day, enabling faster decisions.

Text to Query Generation

What is Text-to-SQL?

Text-to-SQL is the task of converting natural language questions into SQL queries that can be executed on a relational database **

Example:

"Show me total sales by region for the last quarter."

```
SELECT ORDR_RGN, SUM(ORDR_AMT) FROM SLS_ORDR_HDR WHERE quarter =  
'Q4' GROUP BY ORDR_RGN;
```

** - DB agnostic and can run on any DB anywhere including IBM DB2 and watsonx.data

Text to SQL- Onboard data

Prepare and Analyze Metadata:

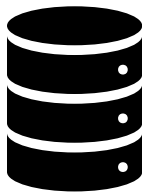
- **Catalog** table and column names with descriptions
- **Expand** cryptic names using domain knowledge and NLP
- **Assign** business concepts to metadata
- Analyze and index **relationships**
- Includes **PK-FK** relationships
- Profile columns and collect **data samples**
- **Vectorize** data samples using Slate 30m embedding model
- Leverage LLMs (Slate 30m) for semantic **metadata vectorization**

Enriched Metadata

Assets (1) Columns (117) Review key relationship				
Find columns (search by column name, asset name, business term, classification, or data class) Last refreshed: 10/7/24, 2:40 PM				
Column	Display name	Description	Business term	
☐ FIC_FIL_NAME	fictitious filing name <a>Suggestion	■ AI The name of the fictitious company as it appears on the fictitious company's filing with the IRS.	• Fictitious Name	
☐ FIC_FIL_PAGES	fictitious filed pages <a>Suggestion	■ AI The number of pages in the fictitious company's filing with the IRS.	— <a>1 suggested	
☐ FIC_FIL_STATE	fictitious filing state <a>Suggestion	■ AI The state in which the fictitious company is located.	— <a>1 suggested	
☐ FIC_FIL_STATUS	fictitious filed status <a>Suggestion	■ AI The status of the fictitious company's filing with the IRS.	— <a>3 suggested	
Items per page: 50 1-50 of 117 items			1 of 3 pages	

Semantic Enrichment

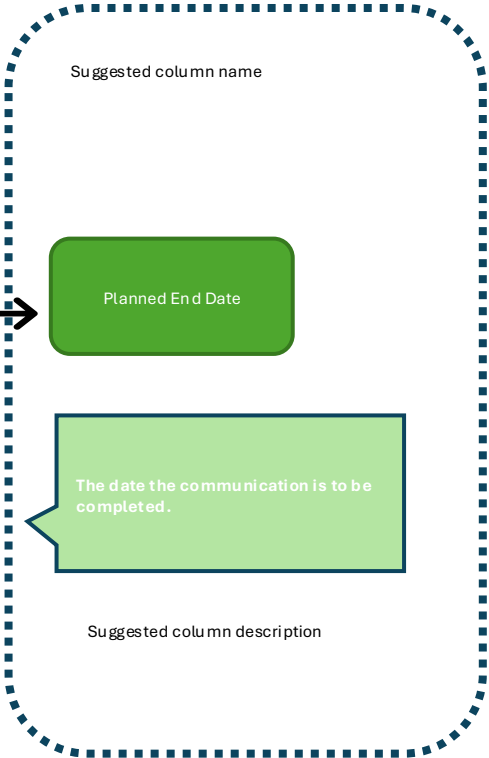
Business Glossary Concepts



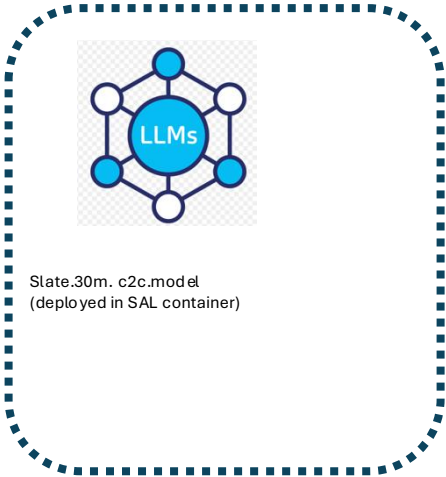
Database Tables

COMMUNICATION

EMPE_D
CMM_ID
PLN_END_DT
ACT_STRT_DT
BSC_DATA_INC OM
VLD_TO_TMS
CST_TXT_ID
....



Enrichment I



Enrichment II

Communication Expected End Date : The date on which the Communication is or was expected to conclude.



OUTPUT



Ranked list of Business Glossary concepts



Metadata Enrichment now with LLMs

What are we introducing?

- **New** - Automatically generate meaningful column names and descriptions with context
- **Enhanced** - Assign terms based on semantic meaning and context
- **Early results** - Internal testing showed **double the number of correct column mappings** (reduction in false positives), and **improved precision** of term assignments with LLMs

Why?

Accelerate data curation through increased accuracy and precision of auto term assignments using AI and LLMs.

IBM Cloud Pak for Data

Search in your workspaces

IBM Cloud Acco... BF

Projects / Bank Transactions Assets /

DB2 Bank Database Asset Enrichment

Default setting Edit enrichment

Assets (150) Columns (989) View relationships

Find assets (search by asset or connection name, path or business term) Last refreshed: Apr 15, 2024, 1:35 PM

Name	Display name	Description	Source	Business term
ACCOUNT_HOLDERS Table	AI Account holders	Table containing data of debit acco...	DB2 Bank... / BANK3	AI 2 Suggest
BANK_CLIENTS Table	Banking clients	AI This table contains client data.	DB2 Bank... / BANK3	Customer +1 more
BANK_CUSTOMERS Table	Banking customers	AI This database table contains custome...	DB2 Bank... / BANK3	Customers
CLIENTS Table	Clients	AI This table contains clients wi...	DB2 Bank... / BANK3	+2 more
CREDIT_CHECK Table	Credit score checks	AI This table contains credit score checks.	DB2 Bank... / BANK3	+2 more
MORTGAGE_APP Table	Mortgage applications	AI This table contains mortgage applications.	DB2 Bank... / BANK3	-
SAVINGS_ACCOUNTS Table	AI Saving accounts data	AI Savings accounts transactional data...	DB2 Bank... / BANK3	Savings acco...
SYSTEMS Table	-	AI Table containing system records.	DB2 Bank... / BANK3	AI 3 Suggest
SYSTEMS_2 Table	-	AI Table containing system records.	DB2 Bank... / BANK3	AI 2 Suggest

10 1 - 10 of 150 items 1 of 15 pages

ACCOUNT_HOLDERS

Data

Go to asset in project

Details Governance Keys

Display name AI Account holders

AI-generated content
Confidence score: 83%

Table containing data of debit account holders ranging from 2020 to today.

Assigned by Generative AI 95%

Source DB2 Bank Connection / BANK3

Asset details

Table type: Table
Columns: 15
Rows: 2,941 (sampled)
Format: application/x-ibm

Asset owner Jon...

Enrichment details

Enrichment options

Profile data | Run basic quality analysis | Assign terms | Expand metadata

Sampling method Moderate

Last enriched Apr 15, 2024, 11:35 AM

Details of job run

AI suggested term assignments

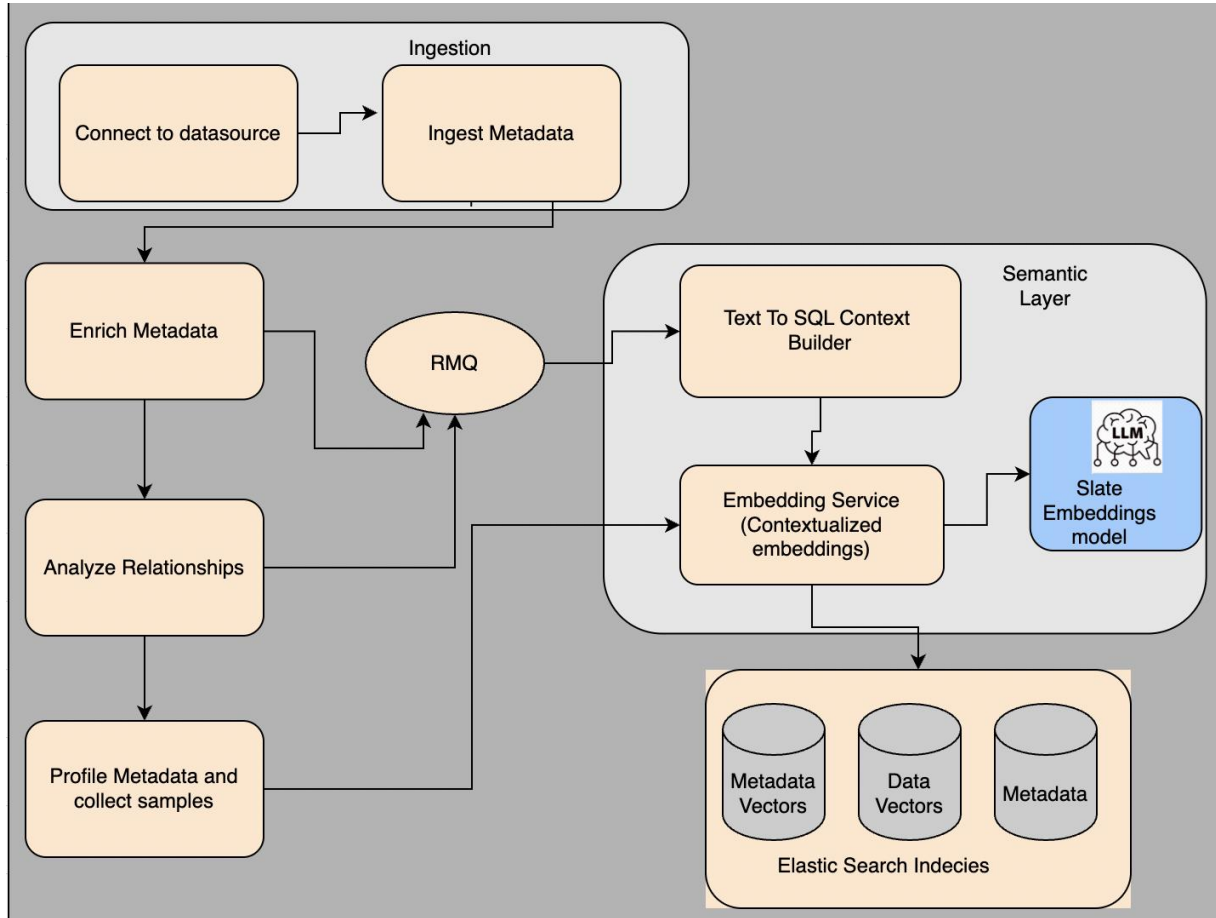
AI-generated description

AI-generated contextual column name

Confidence scores for AI generated content

IBM

Ingest, Enrich and Analyze Metadata

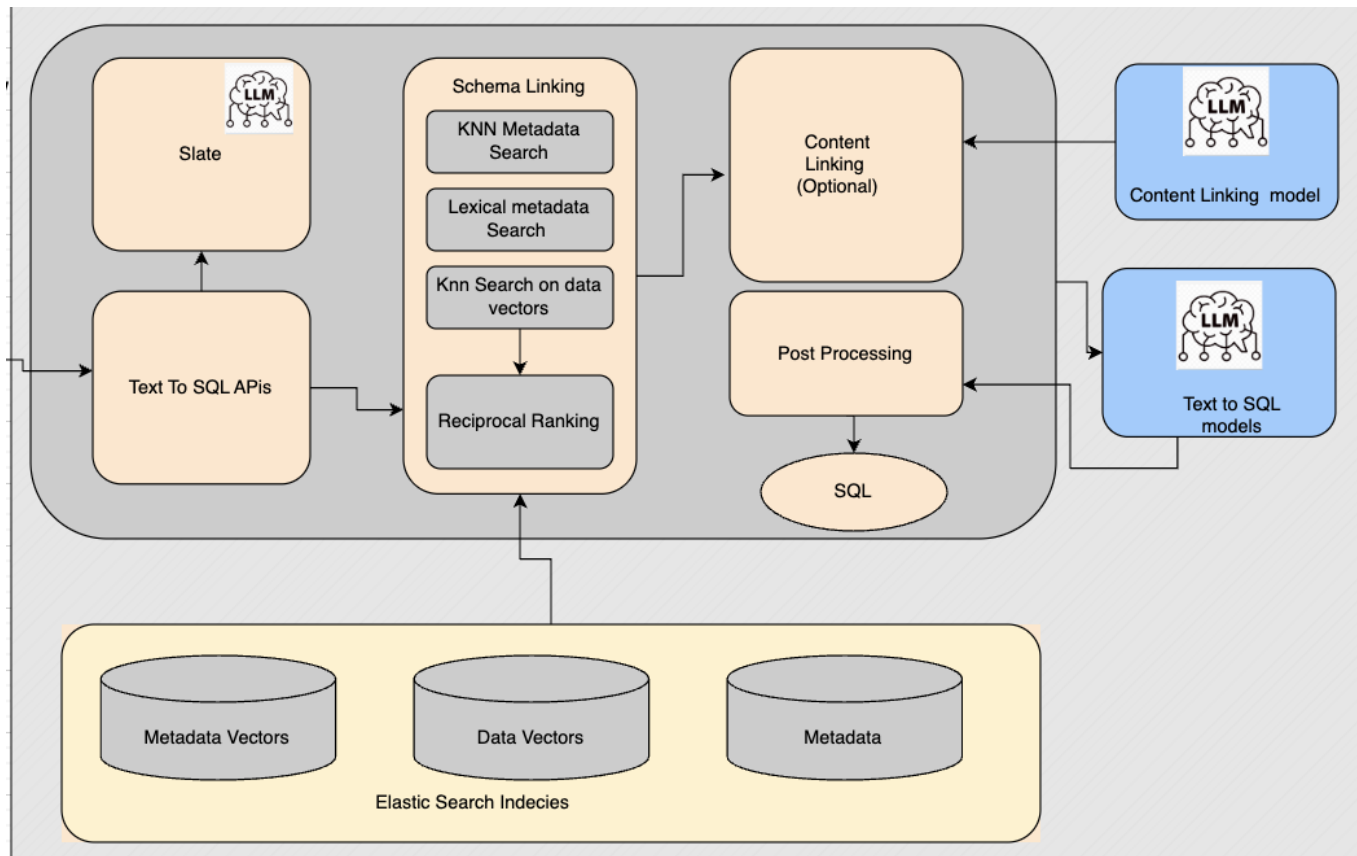


Natural Language to SQL: Retrieval

Schema Linking: Process to identify the relevant columns and tables for the query

- **Convert Natural Language Question to Vectors**
- **Perform Multiple Search Strategies for Accuracy**
 - Metadata vector search
 - Lexical Keyword search
 - Queries on data value embeddingsResults from these searches are **weighted and ranked** to accurately identify relevant columns.
- **Utilize Relationship Information for Context**
- **Generate SQL Using a Hosted Model**

Text To SQL : Retrieval



Example

Request Payload:

```
{
  "query": "How many students of the
Student Club have attended the event
Women's Soccer?",
  "context": {
    "asset_ids": [
      "f02db684-69f5-43bd-bbf4-
53a5edfcd80e"
    ],
    "datalibrary_id": "fe23a89b-5ee1-4b1b-
9d77-6957f980844f"
  },
  "instructions": [
    "Do not use `Student_Club` or `Student
Club` in where clauses"
  ],
  "raw_output": false
}
```

Response :

```
{
  "input_query": "How many students of the
Student Club have attended the event Women's
Soccer?",
  "generated_sql_queries": [
    {
      "sql": "SELECT COUNT(DISTINCT m.member_id)
AS number_of_students FROM student_club.event
e JOIN student_club.attendance a ON e.event_id =
a.link_to_event JOIN student_club.member m ON
a.link_to_member = m.member_id WHERE
LOWER(e.event_name) LIKE '%women%' AND
LOWER(e.event_name) LIKE '%soccer%';",
      "score": 100
    }
  ],
  "model_id": "meta-llama/llama-3-3-70b-instruct",
  "resource_usage": {
  }
}
```

Key Components of Text to SQL

Schema Linking

Custom User
Instructions

Prompt Building

Content Linking

Query Logs

SQL Evaluation and
scoring

Foundation Models used by IBM Data Intelligence from WatsonX.ai

Purpose	Base Model	Status	Comments
Suggest Descriptions Suggest Names (Glossary Generation)	granite-8b-code-instruct	Vanilla model available OOTB in WX.ai on CPD and Saas. Makes it easy to leverage the model. Since IKC is now using a vanilla model with few shots learning, it is easy to provide more context and get better results without having to retrain the model.	Needs 1 GPU. This is a smaller model than previously using custom granite.13b.semantic-automation. The memory foot print is less for the new model than the previous model. The new model generates more accurate names and is used to provide verbose table descriptions.
Text2sql	Llama-3-3-70b-instruct	1)Granite-3-1-8b-instruct (Required) 2)Llama-3-3-70b-instruct (An option for replacing granite model, for better accuracy) 3)slate-30m-english-rtrvr_v2 (Schema linking)	1)Fine tuned Model. deployed on wx.ai as custom model in SAL owned deployment 2)Llama model is available on wx.ai as Foundation model. Non fine tuned 3)Fine tuned Model, deployed on wx.ai as custom model in SAL owned deployment

A Federal Government Agency + IBM

- The current network/system-centric architecture makes the process of retrieving and generating data complex, which impedes decision-making. The goal of the project was to reference data mesh practices and showcase an easier production and search retrieval system of data products for Data Producers and Data Consumers in high importance use cases.
- The federal agency wants to accelerate the adoption of data, analytics, and artificial intelligence from the boardroom to the battlefield that enable decision advantage.



- After initial briefings and solution design workshops, the CE team created PoC of DPH that discovers data products and shows use of Data Contracts.
- This demonstrates Data Product Hub's ability to: 1) Create data products, 2) Search and discover data products and 3) Order data products
- This gives the data producers the ability to quickly search and access data products to surge digital services for rapid response to crises and emergent challenges. In addition, it enables the production needed data products as well.

A Healthcare Insurance Company + IBM

- A healthcare insurance provider is looking to remove duplicate work, optimize costs, and improve the speed and efficiency of its employees.
- Business Information Development Consultants (producers) spend hours writing manual queries and conducting data extracts for numerous data products requests for Operations Managers (consumers). The producers receive 1000+ ad hoc requests per year and the process is highly manual and requires SMEs. There is no ticketing of work, no self-service, no governance, or no business accountability,



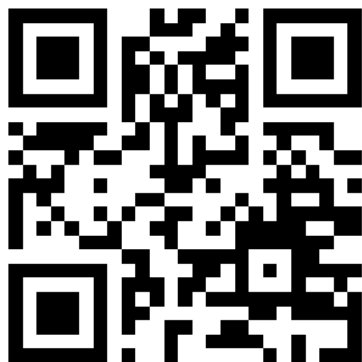
- After solution design workshops, Data Product Hub was introduced where producers can create parameterized queries that can run on top of Snowflake and deliver data extracts in a user-friendly format tailored to the different type of business users across the organization.
- This solution utilizes Data Product Hub to automate the creation data products and increase the reusability of data products.
- The insurance provider can automate their 1000+ ad hoc requests for data products and allow all the business consumers, with no SQL skills, to easily get their data in friendly format.



Speakers



Vijaya Bashyam



Neeru Gupta

